

ILULISSAT JAKOBHAVN, Greenland

WMO: 042210

Lat: 69.240N

Lon: 51.066W

Elev: 31

StdP: 100.95

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.1	-27.1	-34.0	0.2	-28.0	-31.7	0.2	-25.2	16.2	-4.1	14.7	-4.5	1.7	150	0.917

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	15.7	8.2	13.9	7.5	12.2	6.8	9.0	13.2	8.2	11.9	7.6	11.0	7.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
7.1	6.3	8.8	6.2	5.9	8.3	5.5	5.6	8.0	27.0	13.2	25.4	12.1	24.1	11.2	12.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	11.8	10.3	DB	-30.2	18.5	5.3	1.2	-34.0	19.4	-37.1	20.1	-40.0	20.8	-43.9	21.7
			WB	-30.3	10.6	5.1	1.0	-33.9	11.3	-36.9	11.9	-39.7	12.4	-43.4	13.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.5	-11.9	-14.3	-14.1	-7.4	0.2	6.1	8.3	6.7	2.6	-3.4	-6.8	-9.3	(6)
(7)		DBStd	9.43	6.72	7.87	7.66	6.14	3.94	3.14	2.62	2.37	3.10	4.24	5.52	5.45	(7)
(8)		HDD10.0	4966	678	679	747	522	303	124	66	107	223	416	504	597	(8)
(9)		HDD18.3	7983	936	913	1006	772	561	368	310	362	473	673	754	856	(9)
(10)		CDD10.0	23	0	0	0	0	1	6	13	3	1	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.0	5.5	4.7	4.0	3.9	3.4	3.0	2.7	2.7	3.5	4.1	5.4	5.7	(14)
(15)	Precipitation	PrecAvg	258	13	14	14	16	17	24	28	30	35	25	23	19	(15)
(16)		PrecMax	385	44	40	40	46	59	78	79	95	161	98	80	56	(16)
(17)		PrecMin	158	4	1	2	2	2	0	2	0	1	2	1	2	(17)
(18)		PrecStd	58	9	12	10	11	13	19	21	23	33	21	21	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.2	6.5	8.6	13.9	17.1	18.2	16.3	13.6	9.7	8.9	7.9	(19)
(20)			MCWB	2.4	1.2	2.2	3.2	6.5	8.8	9.2	8.7	6.9	3.5	3.7	2.9	(20)
(21)		2%	DB	2.8	2.9	2.1	5.3	10.7	14.8	16.0	14.0	10.8	6.8	6.1	4.5	(21)
(22)			MCWB	-0.7	-1.0	-0.6	1.3	4.9	7.8	8.3	7.6	5.3	2.0	1.6	0.6	(22)
(23)		5%	DB	0.1	0.4	-0.4	3.3	7.5	13.0	14.2	12.1	9.0	3.9	4.1	1.4	(23)
(24)			MCWB	-2.4	-2.6	-2.9	0.1	3.3	6.8	7.7	6.9	4.2	0.9	0.2	-1.3	(24)
(25)		10%	DB	-2.2	-2.3	-3.1	1.0	5.2	11.1	12.6	10.8	6.8	1.9	1.3	-1.1	(25)
(26)			MCWB	-4.4	-4.8	-4.9	-0.8	2.2	6.1	7.2	6.3	3.4	-0.3	-1.2	-3.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	1.6	2.4	3.7	6.9	9.4	10.2	10.0	7.5	4.5	4.2	3.1	(27)
(28)			MCDB	6.0	5.5	5.6	7.5	12.6	15.0	15.7	14.1	12.4	8.0	7.7	6.3	(28)
(29)		2%	WB	-0.4	-0.6	-0.4	1.6	5.2	8.2	9.1	8.6	5.9	2.5	2.0	1.1	(29)
(30)			MCDB	1.8	2.4	2.1	4.2	9.7	13.1	13.8	11.5	9.2	5.6	5.6	4.2	(30)
(31)		5%	WB	-2.5	-2.6	-2.8	0.5	3.8	7.3	8.5	7.8	4.9	1.3	0.5	-1.1	(31)
(32)			MCDB	-0.2	0.3	-0.7	2.7	6.9	11.7	12.4	10.4	7.4	3.5	3.1	1.6	(32)
(33)		10%	WB	-4.5	-4.9	-4.9	-0.9	2.6	6.5	7.9	7.1	4.0	0.1	-1.0	-3.4	(33)
(34)			MCDB	-2.3	-2.4	-3.0	1.0	4.8	10.3	11.3	9.7	6.2	1.6	1.5	-1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	6.3	6.9	6.6	5.2	5.4	5.6	4.8	4.5	5.2	5.2	(35)	
(36)			MCDBR	8.8	9.2	10.2	8.6	8.4	8.9	8.3	8.1	7.0	6.5	7.7	7.6	(36)
(37)			MCWBR	6.9	7.1	8.0	6.2	4.7	4.2	3.9	4.3	4.1	4.5	5.4	5.7	(37)
(38)		5% WB	MCDBR	8.7	9.2	10.0	8.1	7.5	7.6	6.7	6.5	5.8	5.8	7.5	7.7	(38)
(39)			MCWBR	7.1	7.3	8.2	6.0	4.6	4.1	3.8	4.1	3.9	4.5	5.6	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.160	0.210	0.245	0.302	0.312	0.335	0.328	0.321	0.286	0.221	N/A	N/A	(40)
(41)		taud		1.921	2.089	2.279	2.205	2.384	2.423	2.478	2.483	2.482	2.189	N/A	N/A	(41)
(42)		Ebn at Noon		57	573	802	835	877	862	858	820	756	547	N/A	0	(42)
(43)		Edh at Noon		4	40	67	100	96	95	87	77	58	35	N/A	0	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.02	0.36	1.82	3.94	5.61	6.40	5.80	3.89	2.06	0.65	0.06	0.00	(44)
(45)		RadStd		0.00	0.05	0.16	0.38	0.49	0.53	0.56	0.31	0.14	0.05	0.01	0.00	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.87	+3.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.95	+3.68	N/A	+0.50	N/A	N/A	-295	-304	N/A	N/A

Nomenclature: See separate page